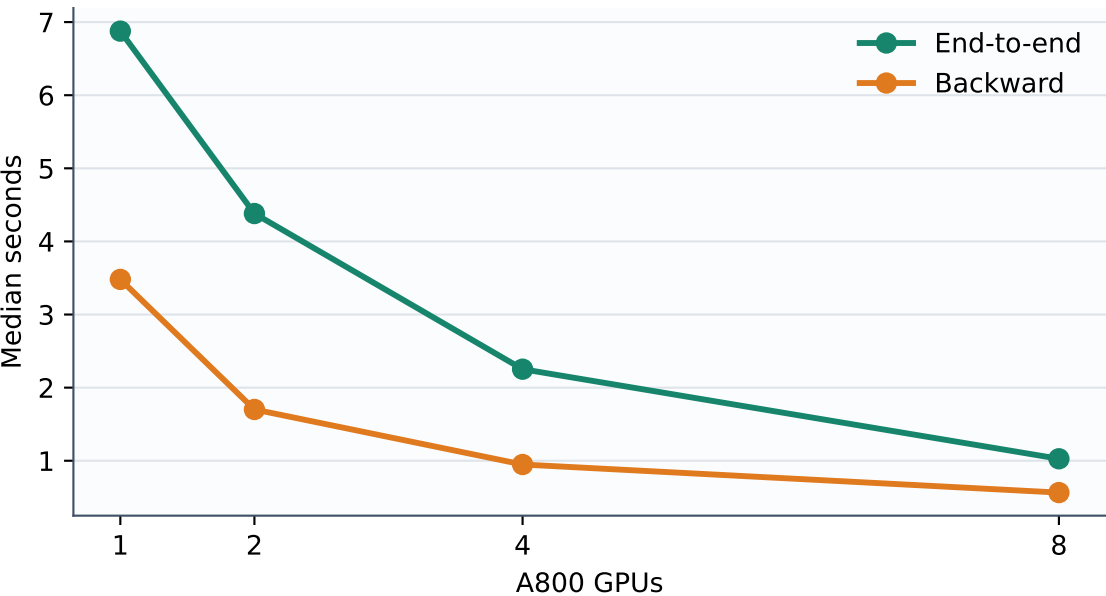
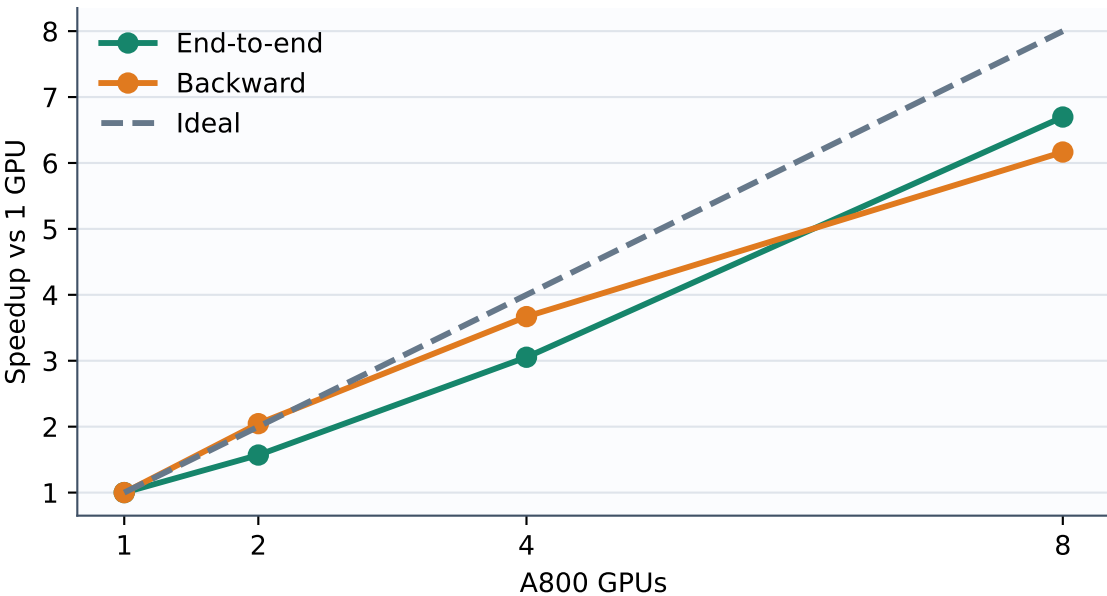


FlagQuantum distributed differentiable statevector · same-host scaling

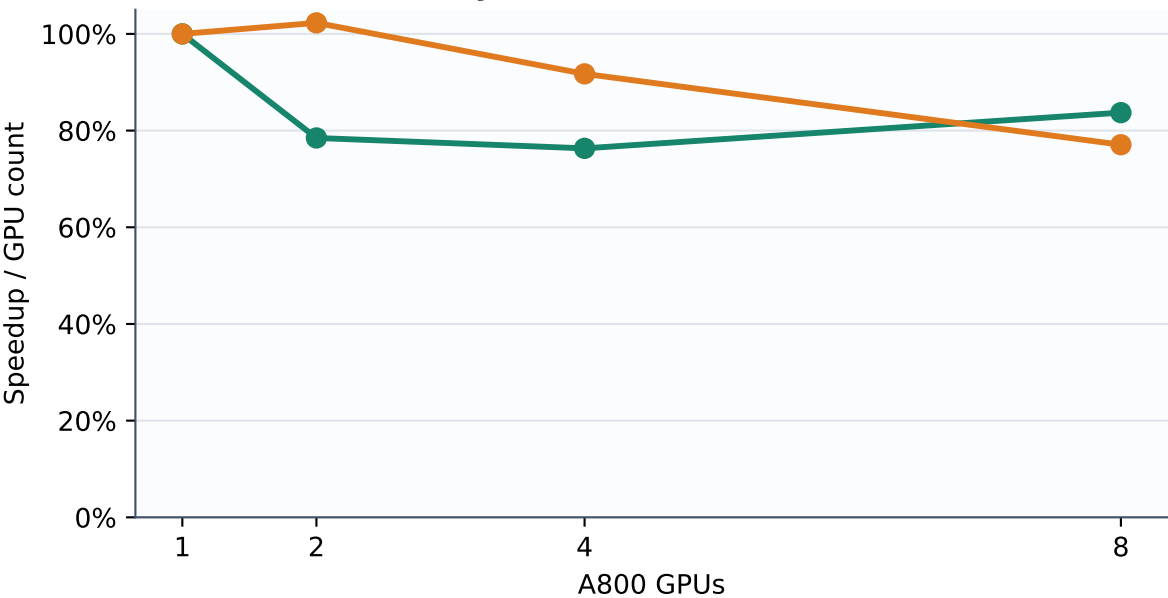
a Fixed-28q training time



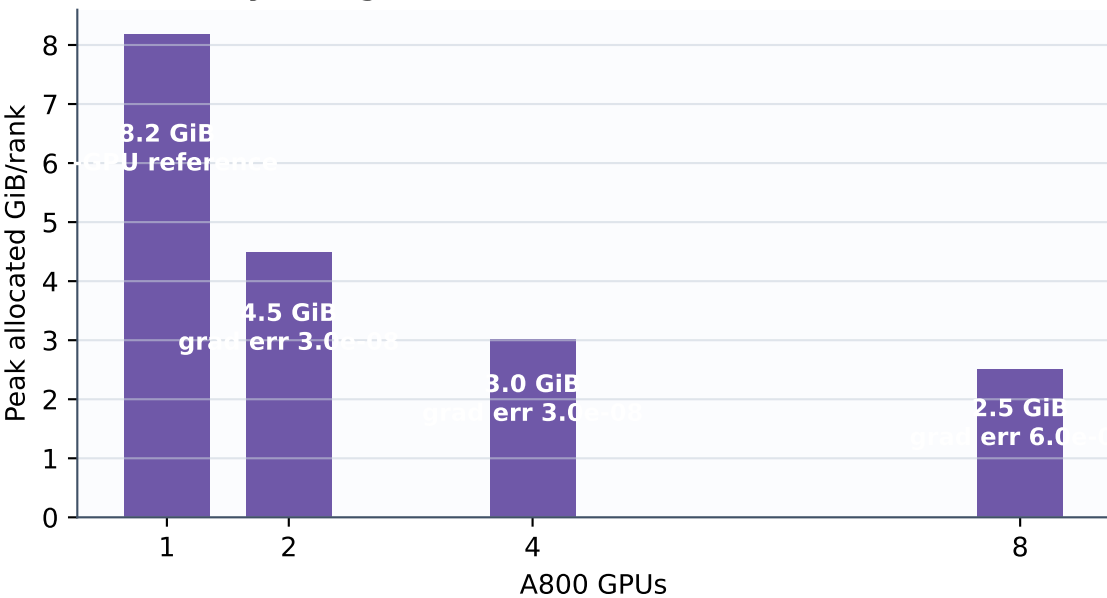
b Strong-scaling speedup



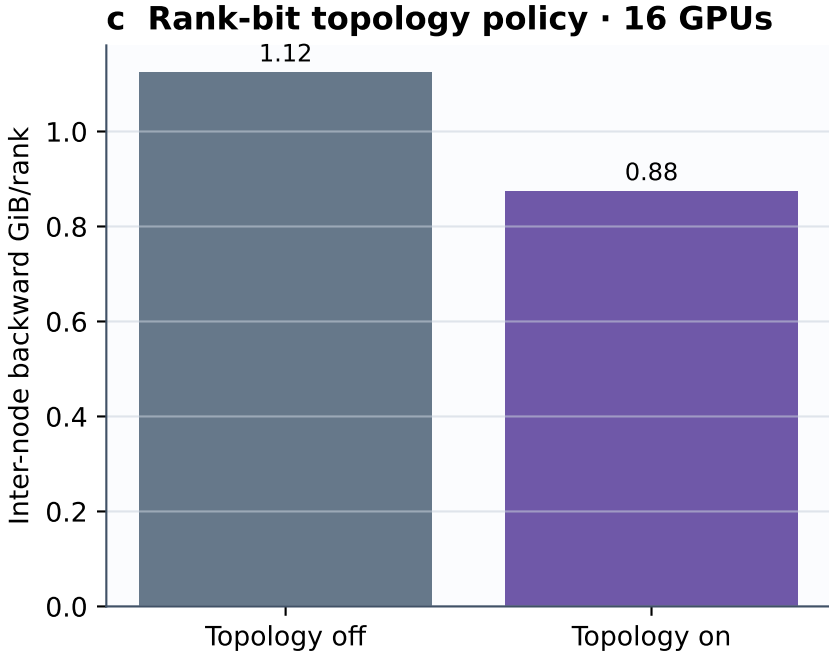
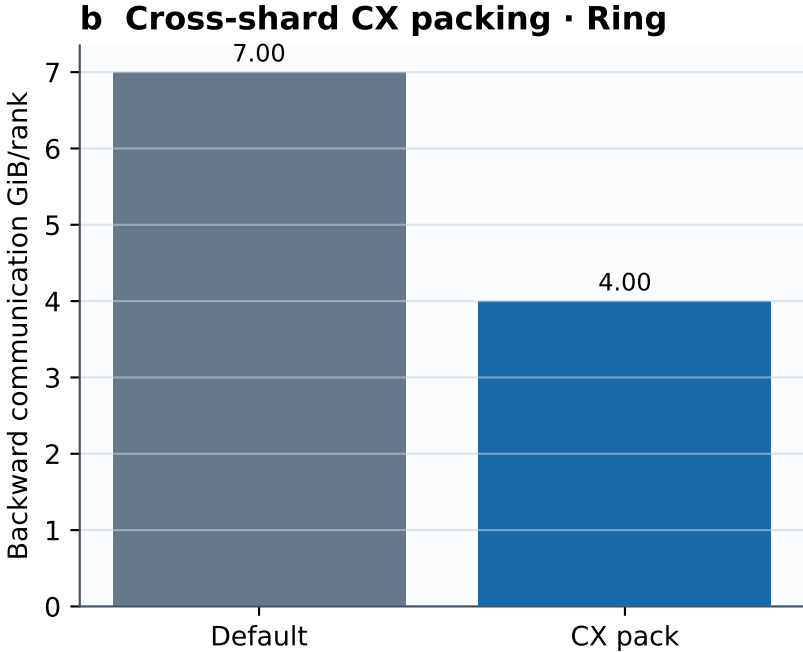
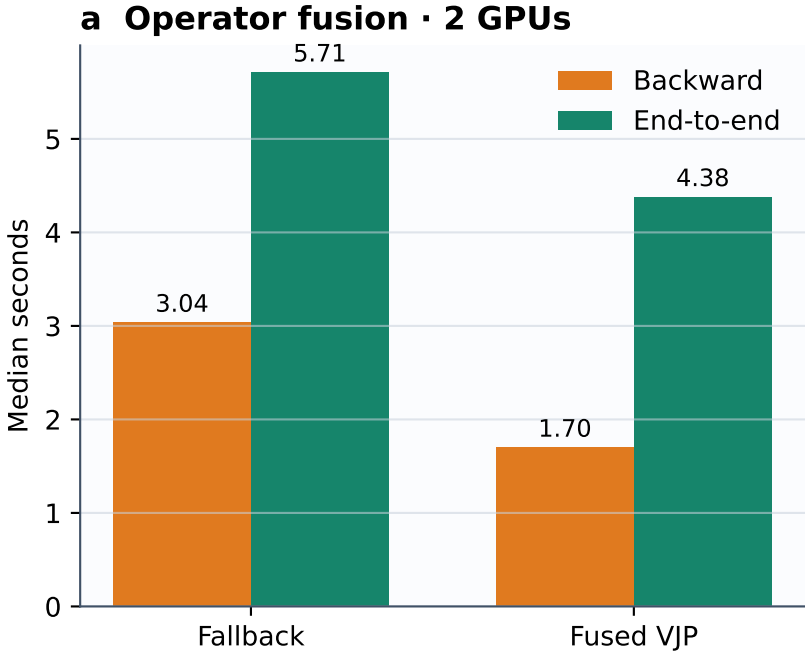
c Parallel efficiency



d Memory and gradient correctness

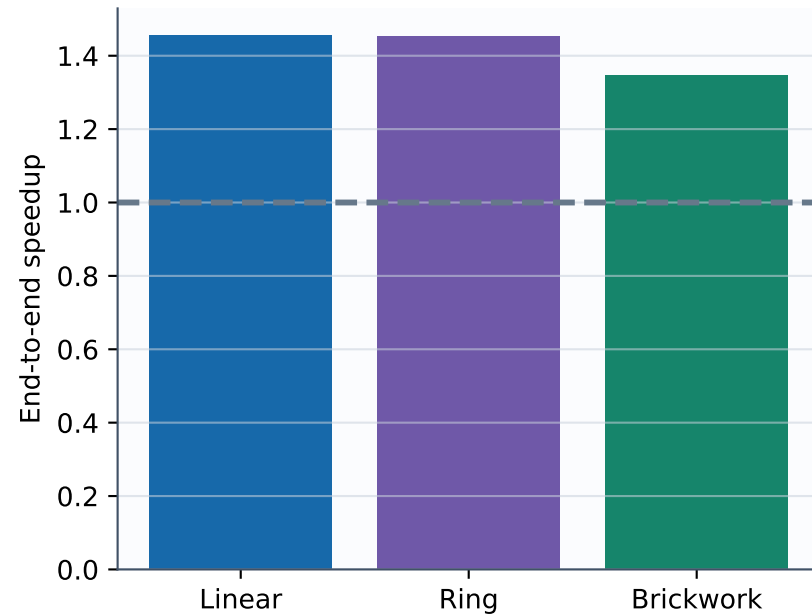


Where the speedup comes from · compute, communication and topology

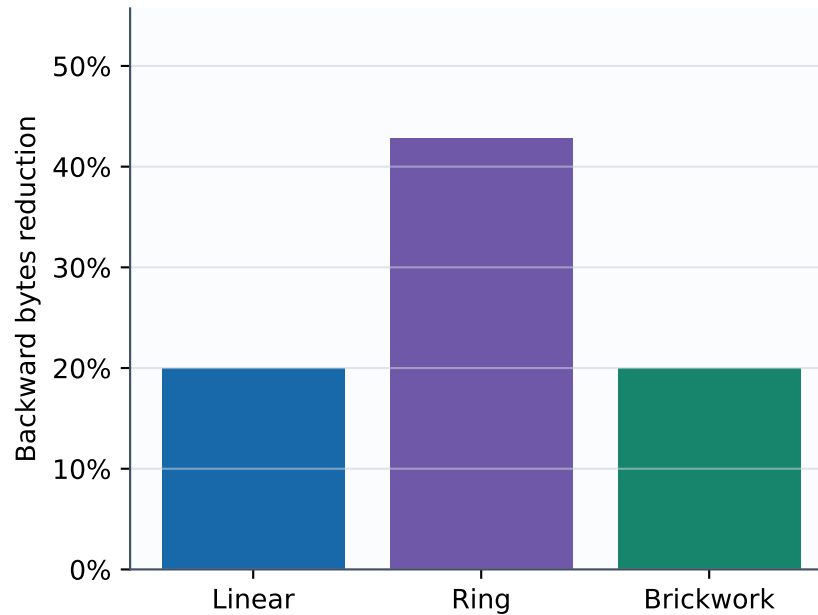


Optimization generality · Full-width Linear, Ring and Brickwork

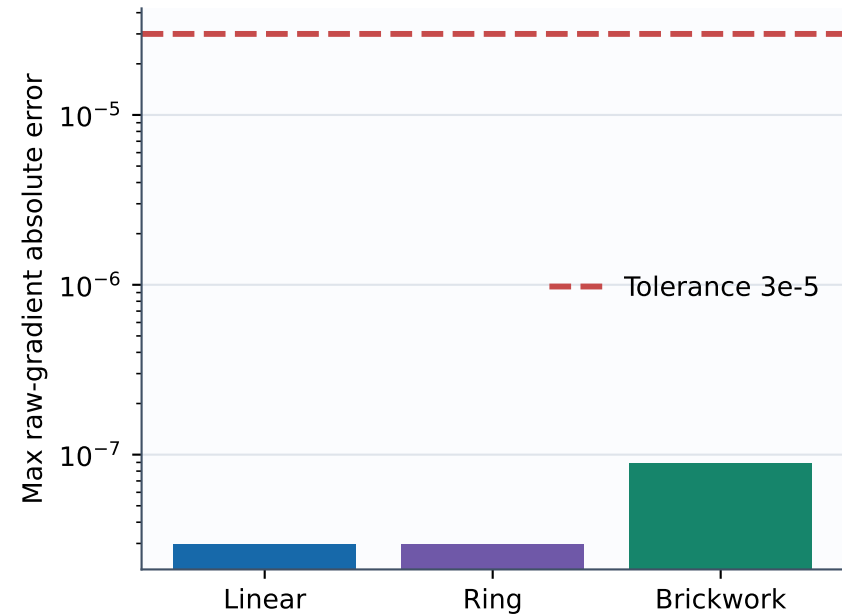
a Improvement across circuit families



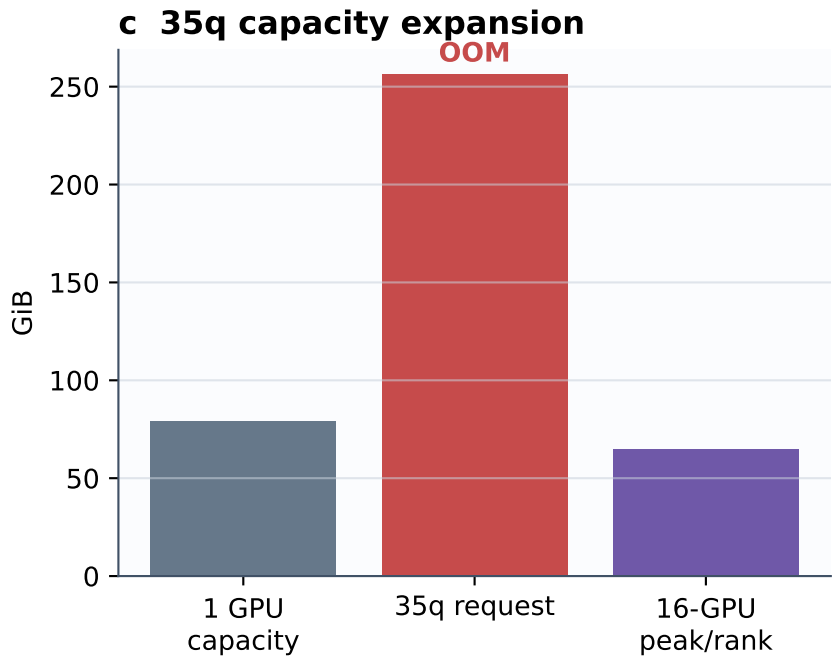
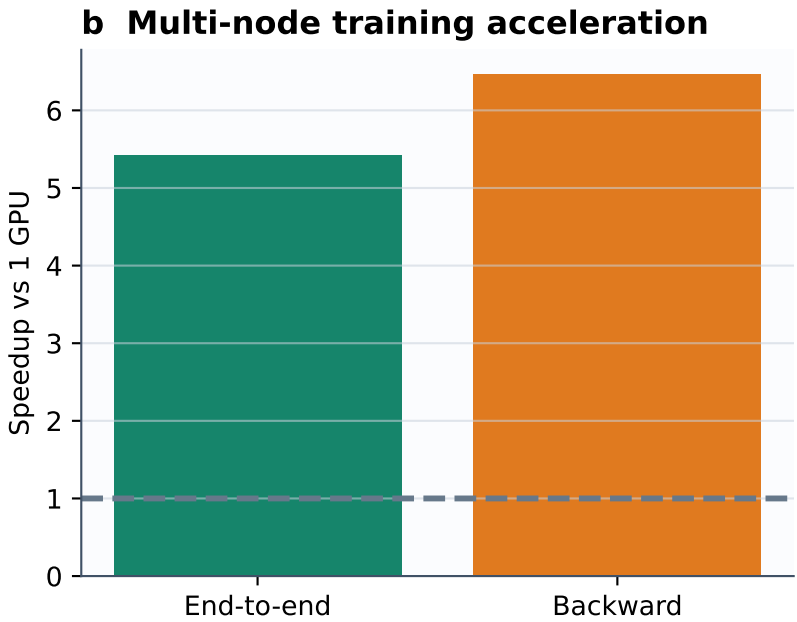
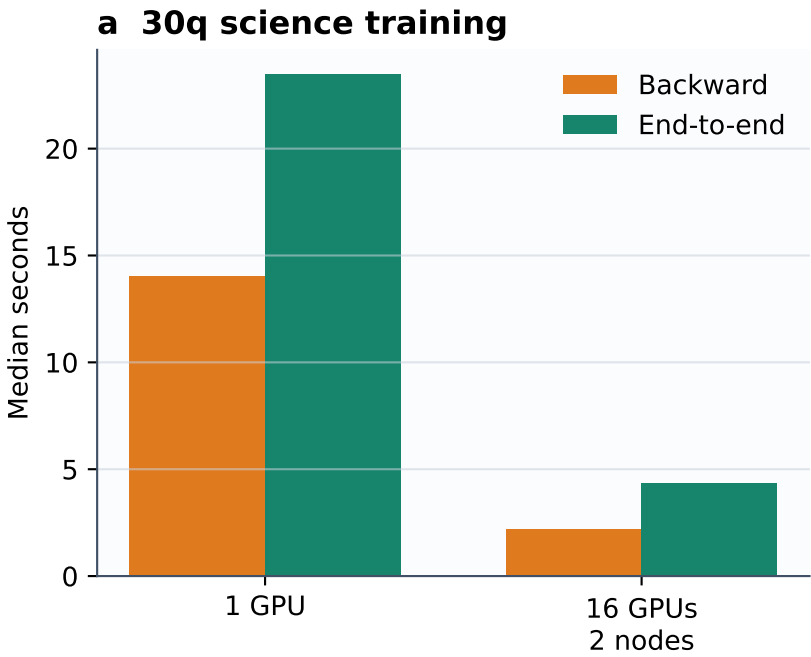
b Communication reduction



c Gradient preservation

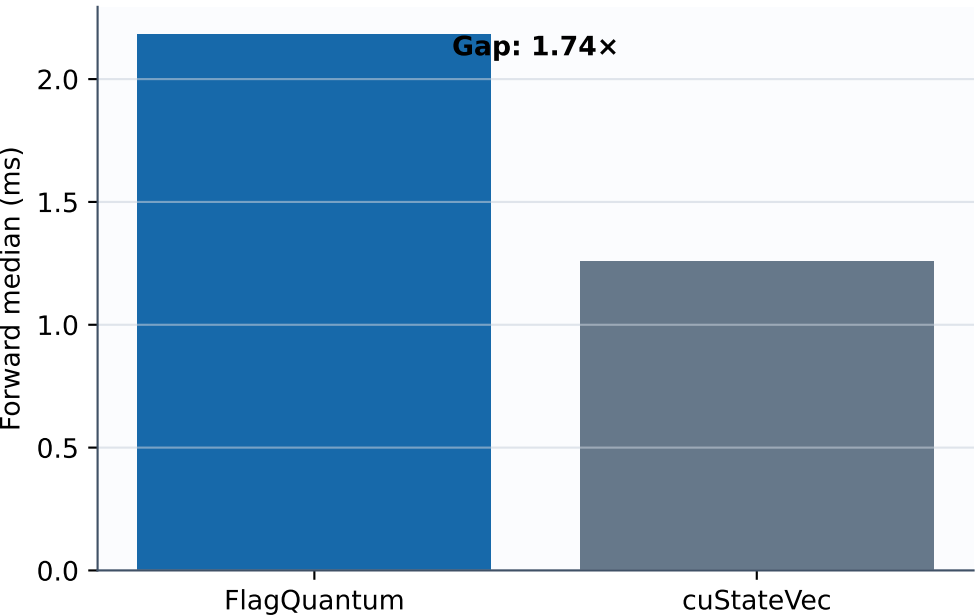


End-to-end result · faster training and a workload that only multi-GPU can run

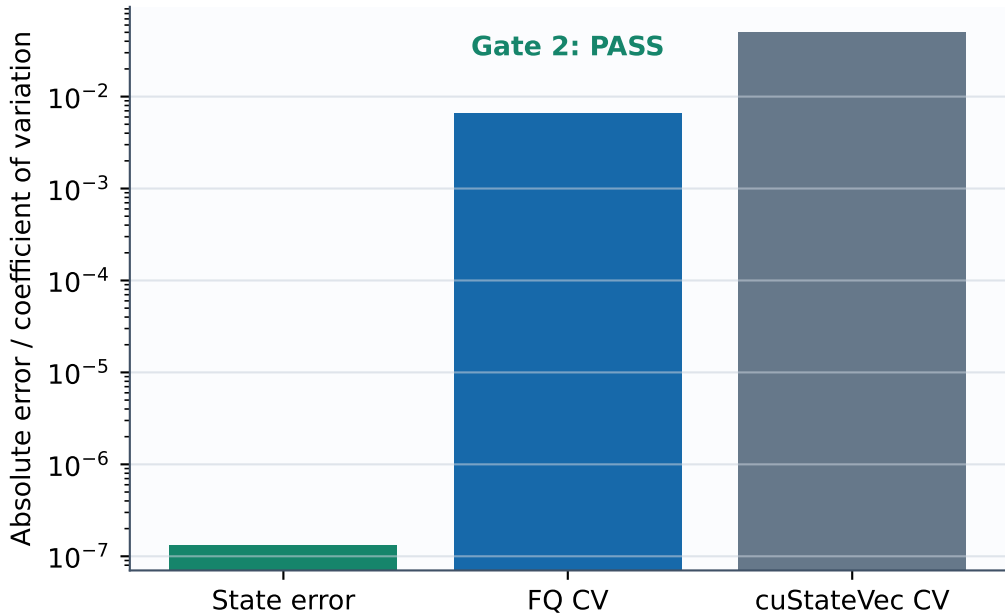


External black-box baseline · benchmark-only, no cuQuantum dependency

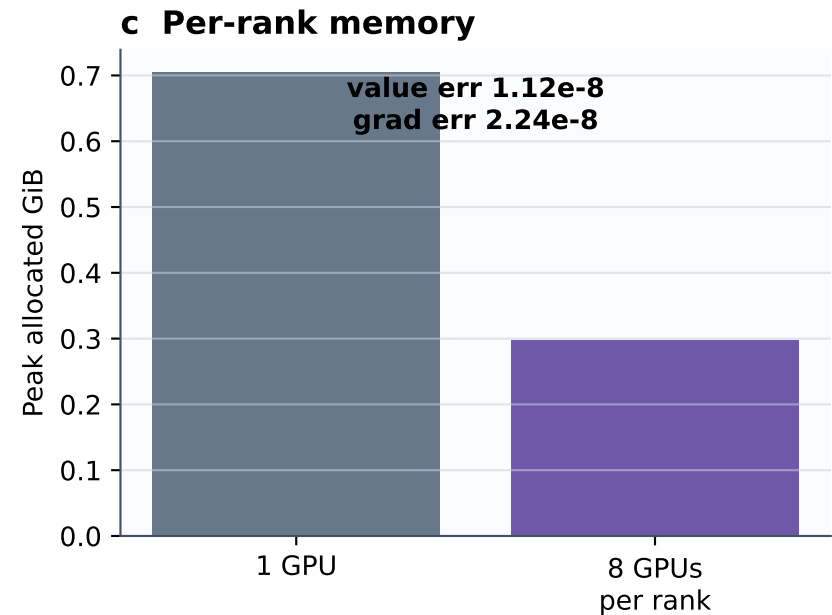
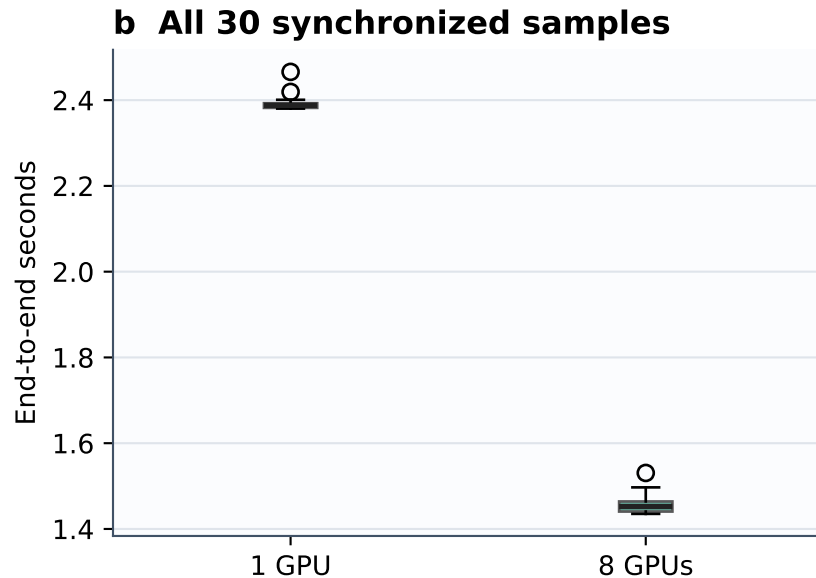
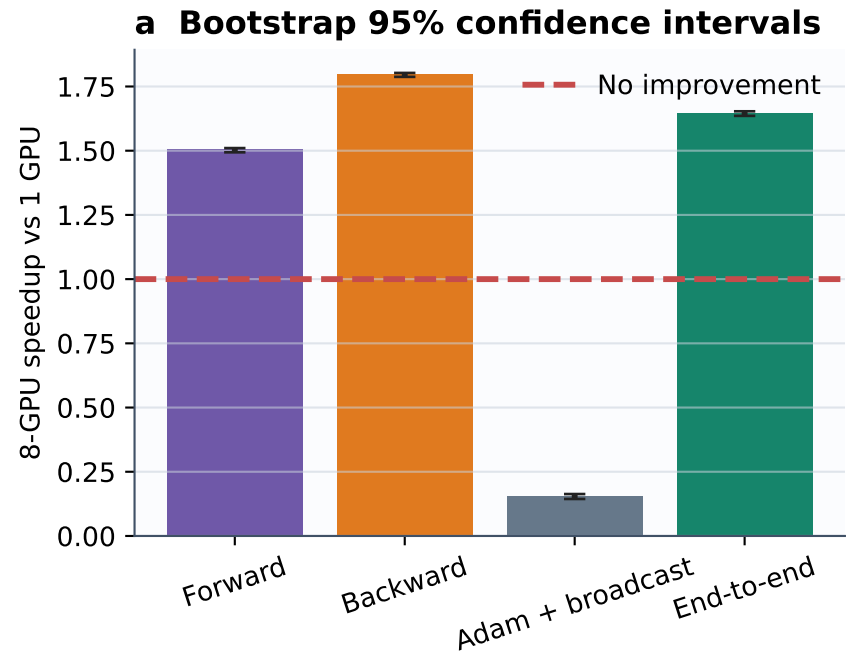
a Same A800, same exact circuit



b Correctness and measurement stability

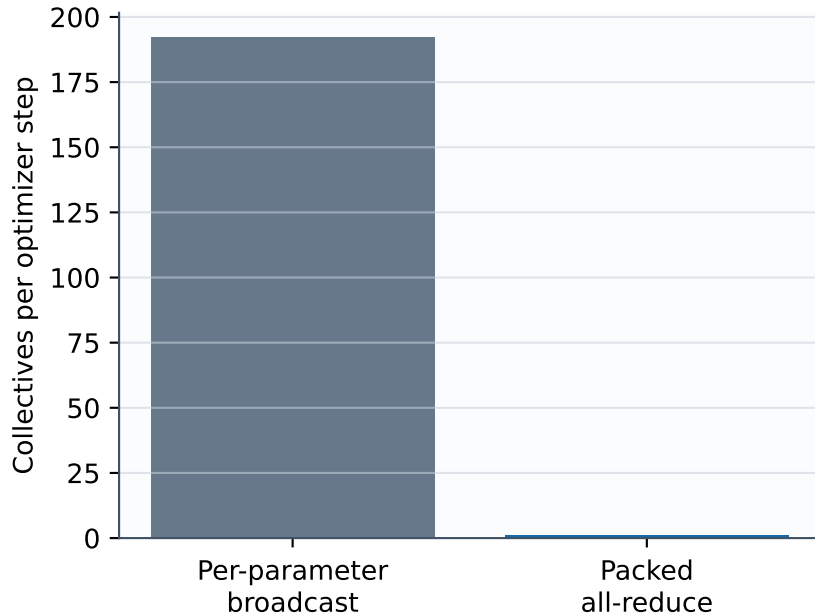


Frozen speed-sweep point · 24q, depth 8, 376 gates, 192 parameters

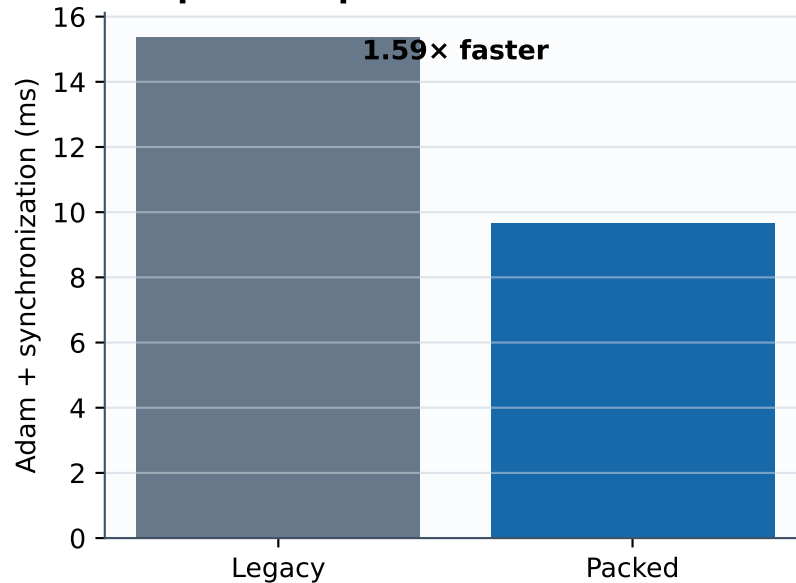


Owner-sharded optimizer synchronization · 192 broadcasts collapsed to one collective

a Collective count



b Optimizer phase



c Full training step

